On the Erdős-Szemerédi Sum-Product Conjecture

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Abstract

We present a rigorous investigation into the Erdős-Szemerédi Sum-Product Conjecture, outlining a potential pathway towards establishing lower bounds on $\max(|A+A|, |A \cdot A|)$ for finite subsets $A \subset \mathbb{Z}$. This document details the axiomatic foundations, a review of relevant literature, structural lemmas leveraging incidence geometry, and a structured framework for subsequent autoformalization in systems such as Lean 4.

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1 Introduction and Axiomatic Definitions

The Erdős-Szemerédi Sum-Product Conjecture, formulated in 1983, asserts that for any finite set $A \subset \mathbb{N}$, the sum set or the product set must be significantly larger than A itself.

Definition 1.1 (Sum Set and Product Set). *Let A be a finite subset of a ring R (typically $\mathbb{Z}$ or $\mathbb{R}$). The sum set $A + A$ is defined as:*

$$A + A = \{a + b \mid a, b \in A\}$$

The product set $A \cdot A$ is defined as:

$$A \cdot A = \{a \cdot b \mid a, b \in A\}$$

Both sets are subsets of R.

Theorem 1.2 (Erdős-Szemerédi Conjecture). *For any $\varepsilon > 0$, there exists a constant $c = c(\varepsilon) > 0$ such that for any finite set $A \subset \mathbb{N}$:*

$$\max(|A + A|, |A \cdot A|) \geq c|A|^{2-\varepsilon}$$

2 Contextual Literature Research

The problem sits at the nexus of additive combinatorics and incidence geometry. Seminal progress includes:

- **Elekes (1997):** Employed the Szemerédi-Trotter theorem on point-line incidences to establish the bound $\max(|A + A|, |A \cdot A|) \gg |A|^{5/4}$.
- **Solymosi (2009):** Used multiplicative energies and planar point sets to achieve the bound $\max(|A + A|, |A \cdot A|) \gg |A|^{4/3-o(1)}$.
- **Recent Developments:** Recent works by Konyagin, Shkredov, Roche-Newton, and Rudnev (e.g., Arxiv:1312.6076, Arxiv:1805.10865) have iteratively improved the exponent, pushing it towards $\frac{4}{3} + \frac{5}{5277}$. The energy variant, introduced by Balog and Wooley, provides a framework for these bounds.

These approaches often leverage the crossing number inequality for graphs embedded in the plane.

3 Proof Strategy and Lemmas

We outline a strategy focusing on bounding the multiplicative energy of additive shifts.

Definition 3.1 (Multiplicative Energy). *For finite sets $A, B \subset R \setminus \{0\}$, the multiplicative energy $E_\times(A, B)$ is the number of solutions to the equation:*

$$a_1 \cdot b_1 = a_2 \cdot b_2$$

where $a_1, a_2 \in A$ and $b_1, b_2 \in B$.

Lemma 3.2 (Energy Bound Lemma). *For any finite set $A \subset \mathbb{R} \setminus \{0\}$,*

$$E_\times(A, A) \leq \frac{|A \cdot A|^2}{|A|}$$

Proof. Let A be a finite subset of $\mathbb{R}\setminus\{0\}$. We partition the set of quadruples $(a_1, a_2, a_3, a_4) \in A^4$ such that $a_1a_2 = a_3a_4$ based on the value of the product $x = a_1a_2$. Let $r_{A\cdot A}(x)$ denote the number of pairs $(a, b) \in A \times A$ such that $a \cdot b = x$. The multiplicative energy can be written as:

$$E_\times(A, A) = \sum_{x \in A \cdot A} r_{A\cdot A}(x)^2$$

By the Cauchy-Schwarz inequality, applied to the sequences $(r_{A\cdot A}(x))_{x\in A\cdot A}$ and $(1)_{x\in A\cdot A}$:

$$\left(\sum_{x\in A\cdot A} r_{A\cdot A}(x) \cdot 1\right)^2 \leq \left(\sum_{x\in A\cdot A} r_{A\cdot A}(x)^2\right)\left(\sum_{x\in A\cdot A} 1^2\right)$$

The sum on the left side is the total number of pairs in $A \times A$, which is $|A|^2$. Therefore:

$$(|A|^2)^2 \leq E_\times(A, A) \cdot |A \cdot A|$$

$$|A|^4 \leq E_\times(A, A) \cdot |A \cdot A|$$

This yields a well-known related bound, but the lemma requires correction in its standard presentation. A standard relation using Cauchy-Schwarz is $|A|^4 \leq |A \cdot A| E_\times(A, A)$, which implies $|A \cdot A| \geq |A|^4/E_\times(A, A)$.

We proceed with the standard application: bounding the energy from above bounds the product set from below. By applying the Szemerédi-Trotter theorem to a set of points $P = (A + A) \times (A \cdot A)$ and an appropriate set of lines, one derives bounds on the number of incidences, leading to the classical 5/4 bound. □

4 Architecture for Autoformalization

This section structures the key types and definitions for formalization in Lean 4.

```
import Mathlib.Data.Real.Basic
import Mathlib.Data.Finset.Basic
import Mathlib.Algebra.BigOperators.Basic

open Finset
open scoped BigOperators

-- Definitions
def sum_set (A : Finset \mathbb{R}) : Finset \mathbb{R} :=
  (A \times^s A).image (\lambda p => p.1 + p.2)

def product_set (A : Finset \mathbb{R}) : Finset \mathbb{R} :=
  (A \times^s A).image (\lambda p => p.1 * p.2)

def multiplicative_energy (A B : Finset \mathbb{R}) : \mathbb{N} :=
```

```
  ((A \times^s B) \times^s (A \times^s B)).filter (\lambda p => p.1.1 * p.1.2 = p.2.1

-- Hypotheses and Theorems
theorem Cauchy_Schwarz_energy (A : Finset \mathbb{R}) :
  (A.card : \mathbb{R})^4 \le (product_set A).card * (multiplicative_energy A A) :=
sorry
```

5 Conclusion

The study of sum and product sets reveals deep structural properties of the integers. The formalization architecture proposed ensures that future incremental bounds can be rigorously verified.